Fourier Transform

Audio Tech I
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Monday, November 17

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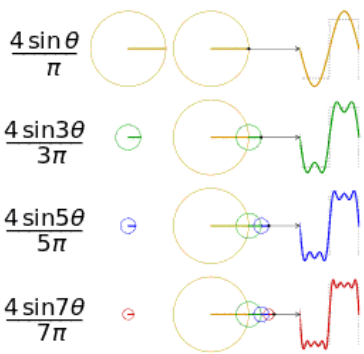
Lesson Outline

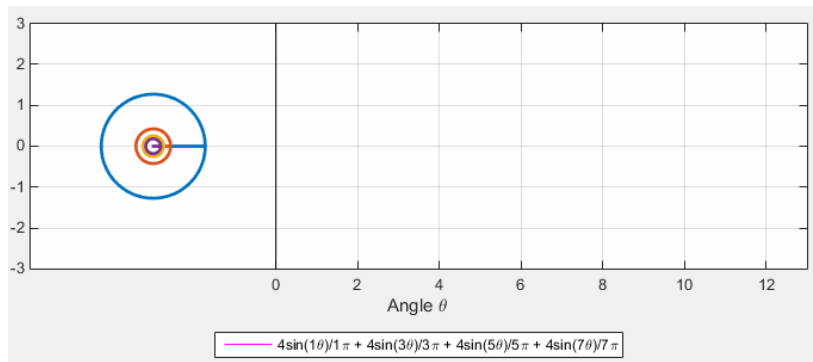
- Fourier Transform
- Discrete Fourier Transform
- Up Next…

Fourier Series

$$f(t) = \frac{1}{2}a_0 + \sum_{k=1}^{\infty}(a_k \cos(2\pi kt) + b_k \sin(2\pi kt))$$

A **periodic** waveform can be represented by a (infinite) sum of sinusoids at frequencies that are multiples of the fundamental





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Fourier Transform

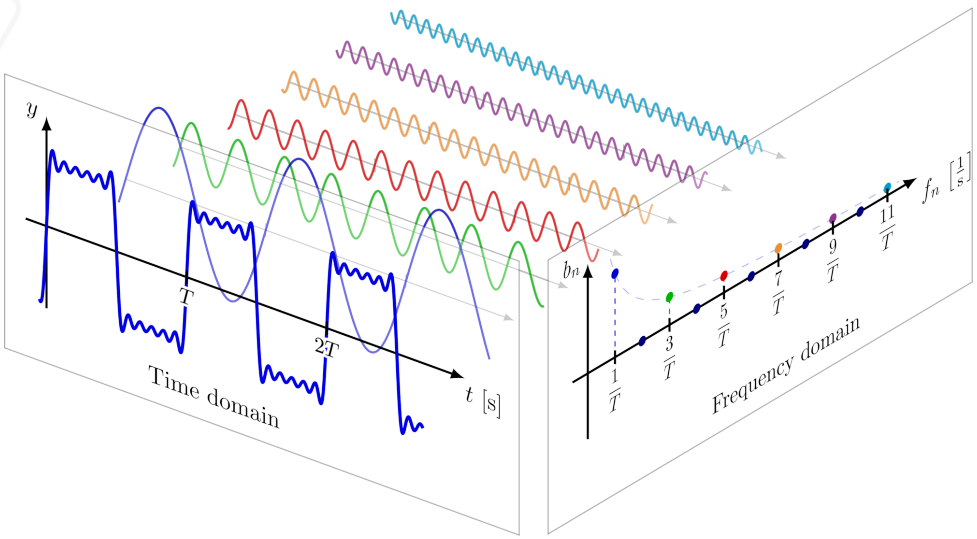
- Moves our signal from the time domain to the frequency domain
  - Inverse Fourier transform takes it back

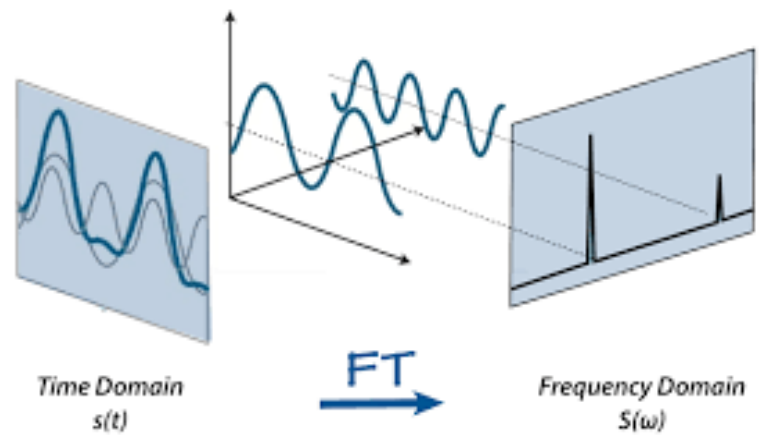
$$x(t) \rightarrow X(\omega)$$

- Amplitudes over time into frequencies at respective amplitudes (and phases)
- Gives us a “recipe” for what frequencies make up our signal

Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi\omega t}dt$$





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What does this actually say???

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi\omega t} dt$$

The (infinite) sum of our signal correlated with an analyzing signal (for all frequencies)

↓

a single frequency signal

Fourier Transform

“Taste Filters”

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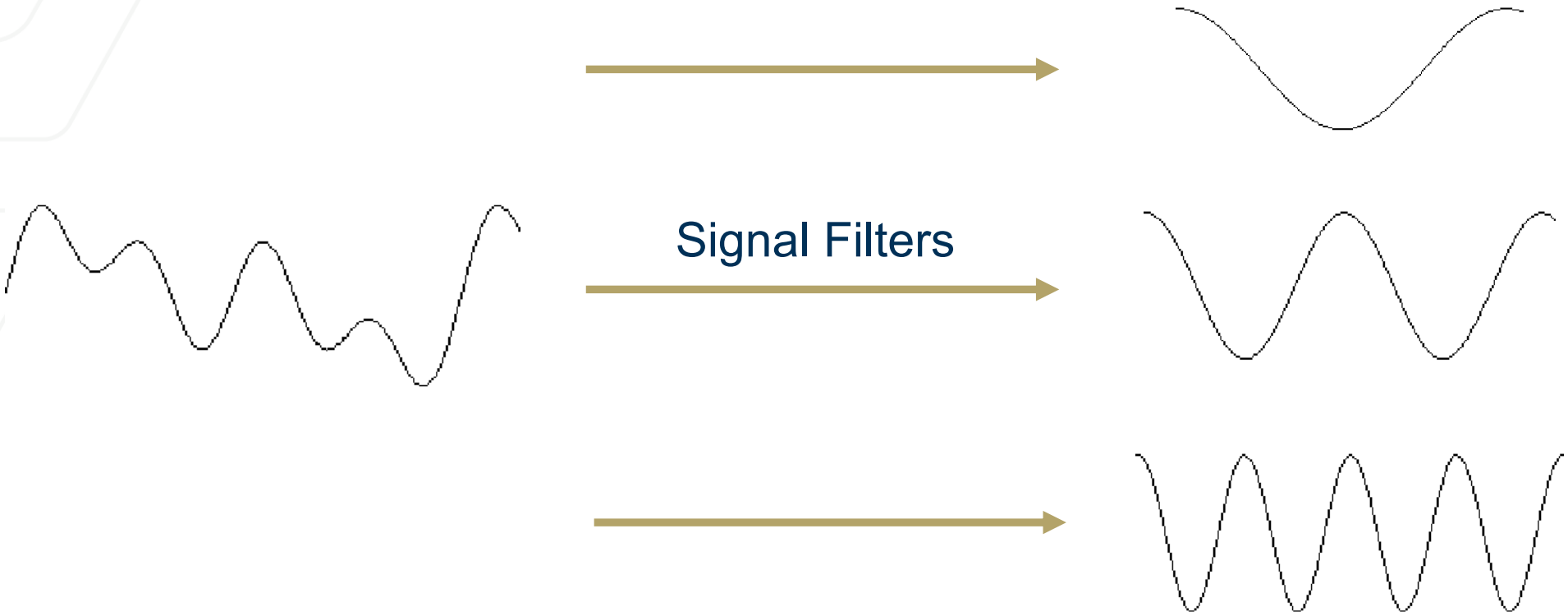
Fourier Transform



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Discrete Fourier Transform

Discrete Fourier Transform

$$X_k = \sum_{n=0}^{N-1} x_n e^{\frac{-j2\pi}{N}kn}$$

- In the digital world, we don't have infinite time or infinite frequency
- Instead of looking at all possible frequencies and time
  - We'll look for frequencies from 0 Hz to the sampling rate
  - Within a discrete number of samples

Discrete Fourier Transform (DFT)

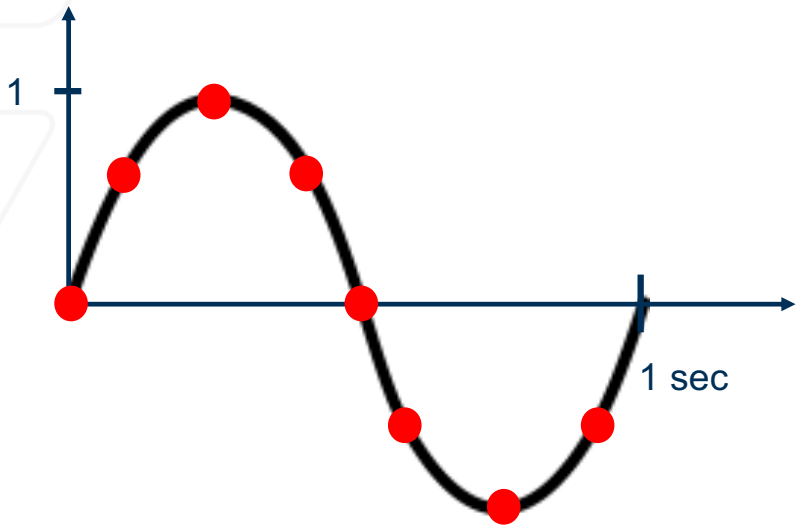
$$X_k = \sum_{n=0}^{N-1} x_n e^{\frac{-j2\pi}{N}kn}$$

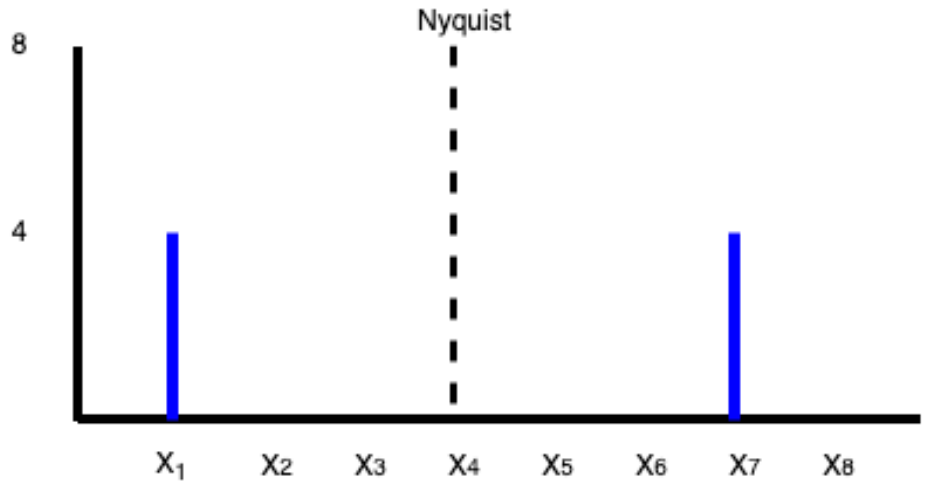
Finding the amplitude of each frequency bin, k:

- From 0 to the number of samples
- our signal (in samples) correlated with an analyzing signal (at each frequency bin)

DFT “perfect example”

- If I have a 1Hz sine wave sampled at 8Hz…





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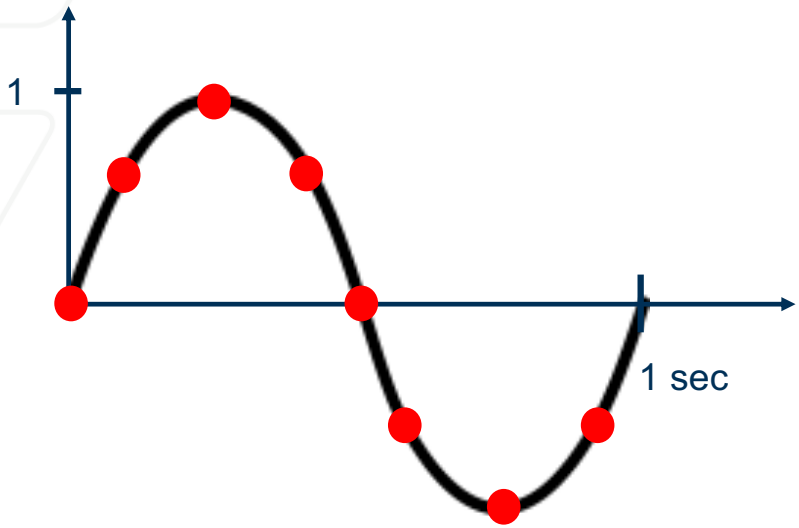

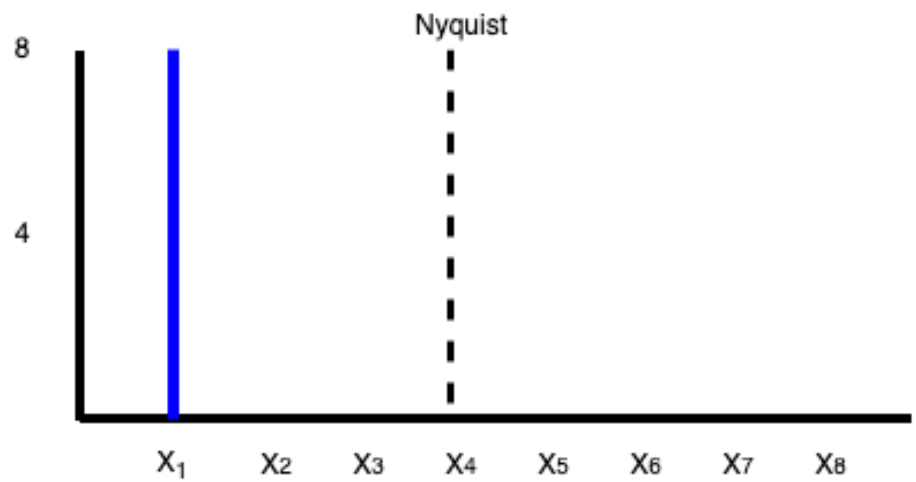

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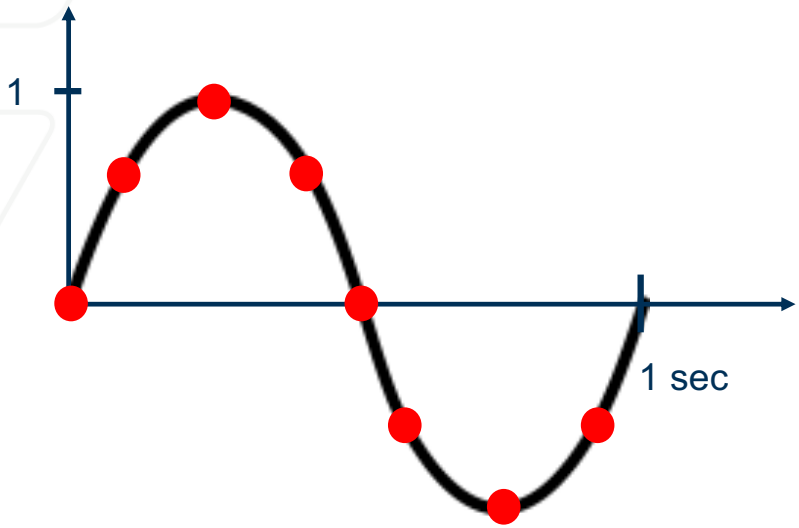
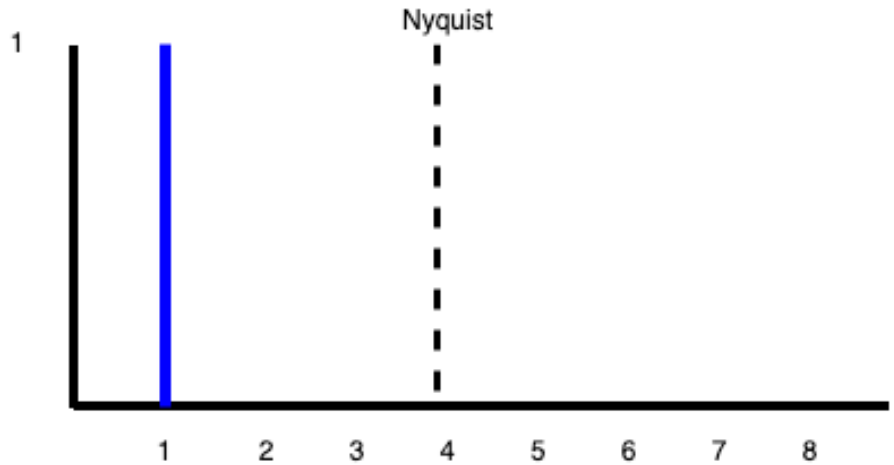




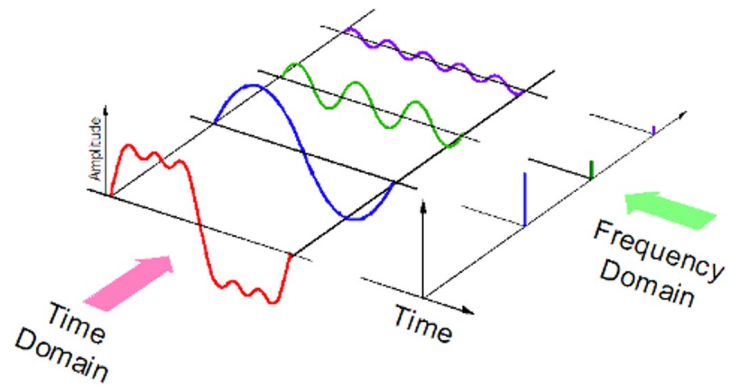
Real World Issues

- Signals in the real world are not periodic
- Taking the DFT of a 3-minute song is not very helpful
- DFT takes a while to compute (n^2)

Fast Fourier Transform and Short Time Fourier Transform

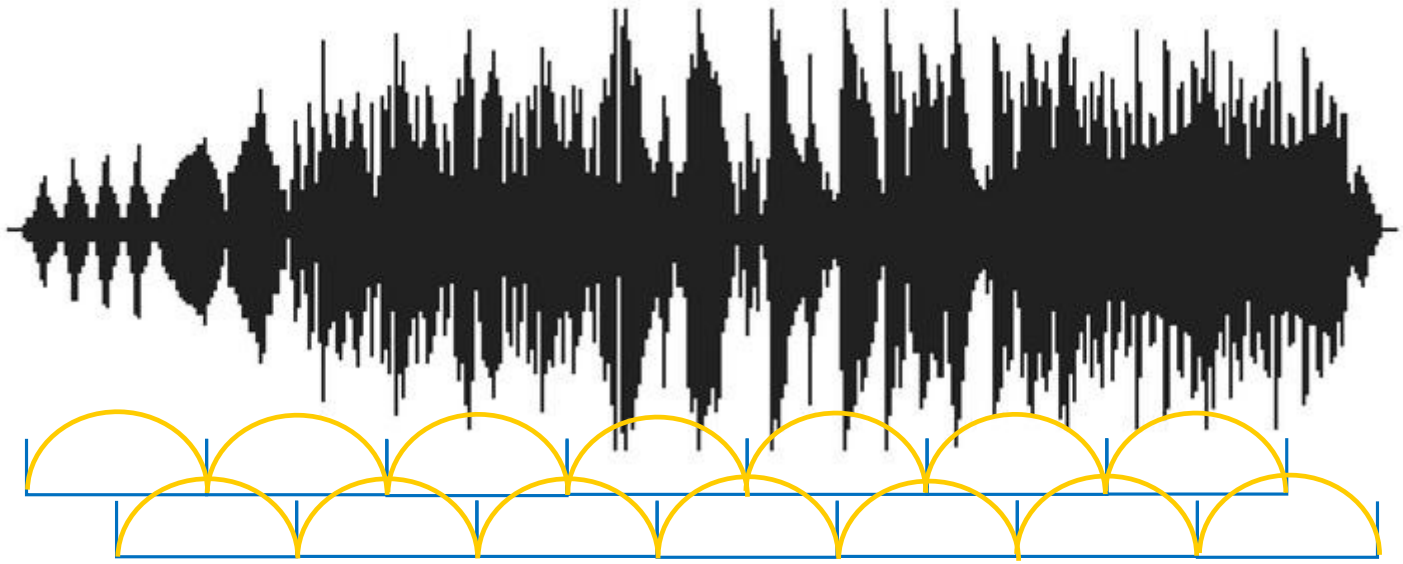
Windowing, Periodization, FFT, STFT

- Windowing
  - Take tiny snippets of the signal and smooth the amplitudes
- Periodization
  - Pretend the signal is periodic
- Fast Fourier Transform
  - The DFT but with nlog(n) runtime
- Short Time Fourier Transform
  - Carrying out multiple FFTs



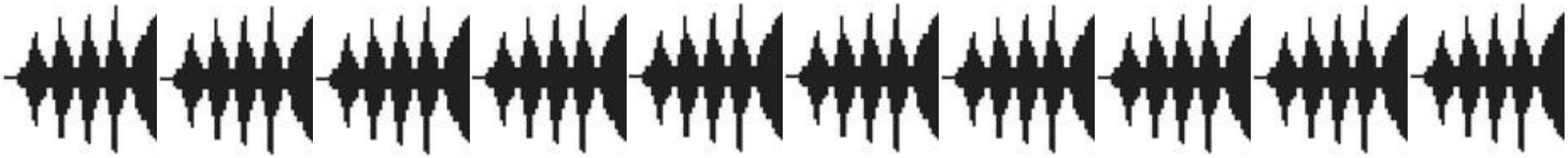
Windowing

- Divide the audio signal into equal size, overlapping frames



Periodization

- Pretend that each frame is part of a periodic function



Transform Output

- Remember the FFT (and the DFT) output complex numbers
- So, you can derive the amplitude and phase of each frequency bin by converting the cartesian real + imag to polar form

FFT

- Input: the original signal, amplitude samples over time (in frame)
- Output: amplitudes and phases of each frequency bin

Number of frequency bins = window size / 2

Frequency bin width = Nyquist frequency / num bins

Windowing and Frequency Bins Example

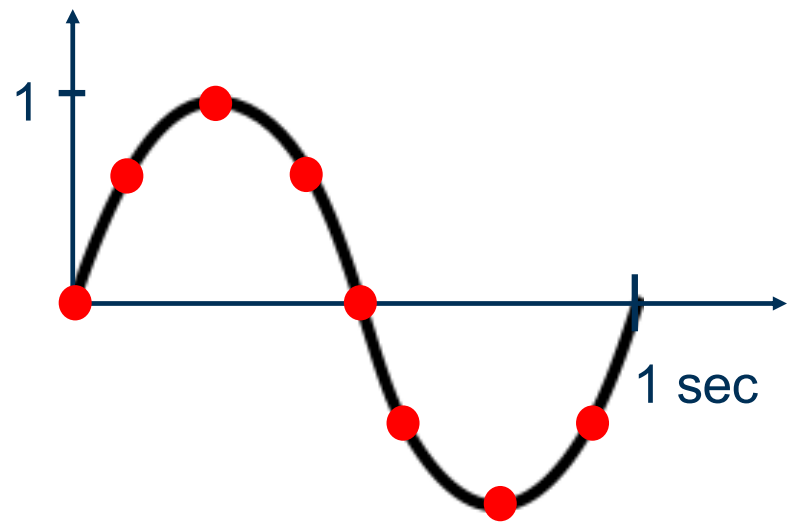
Frame Size = 8 samples

Sampling Rate = 8 Hz

Nyquist Frequency = 8/2 = 4 Hz

Num of Freq Bins = 8 samples / 2 = 4 bins

Freq Bin Width = 4 Hz / 4 bins = 1 Hz per bin



Windowing and Frequency Bins Example

Frame Size = 1024 samples

Sampling Rate = 44.1 kHz

Nyquist Frequency = 44,100/2 = 22,050 Hz

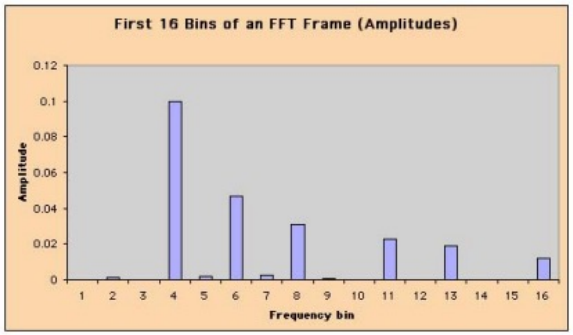
Num of Freq Bins = 1024 samples / 2 = 512 bins

Freq Bin Width = 22,050 Hz / 512 bins = 43 Hz per bin

0 - 43 Hz	43 Hz	86 Hz	129 Hz	. . .	21,921 Hz	21,964 Hz	22,007 Hz	22,050 Hz

Issues and Tradeoffs

- Window Size vs Frequency Bins
  - As we increase window size, we get more frequency resolution but decrease time resolution
  - As we decrease window size, we get better time resolution but decrease frequency resolution
- Lossy Process
- Linear Frequency Bins



Bin Frequency(hz)	Amplitude	Phase
0-22	0.00003	
22-	0.0015	
44-	0.0001	
66-	0.1	
88-	0.002	
110-	0.047	
132-	0.0023	
154-	0.031	
176-	0.0005	
198-	0.00026	
220-	0.023	
242-	0.0001	
264-	0.019	
286-	0.00013	
318-	0.00005	
340-	0.0123	

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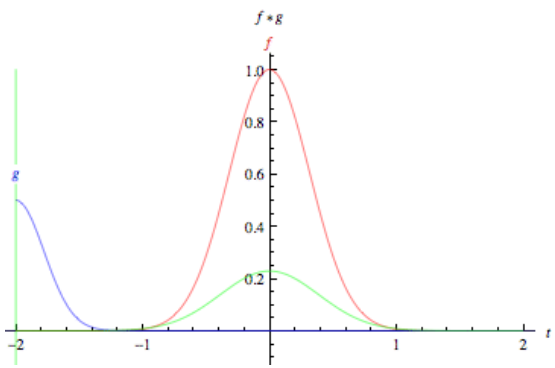
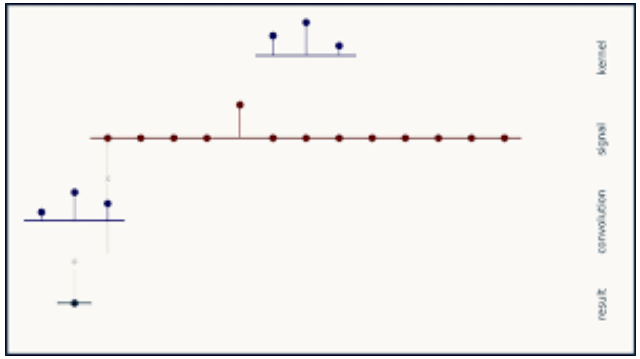
Convolution

- Expresses how the “shape” of one function modifies the other
  - An integral that expresses the amount of overlap of one function as its shifted over another
  - ”Blends” two functions
- $(f * g)(t)$

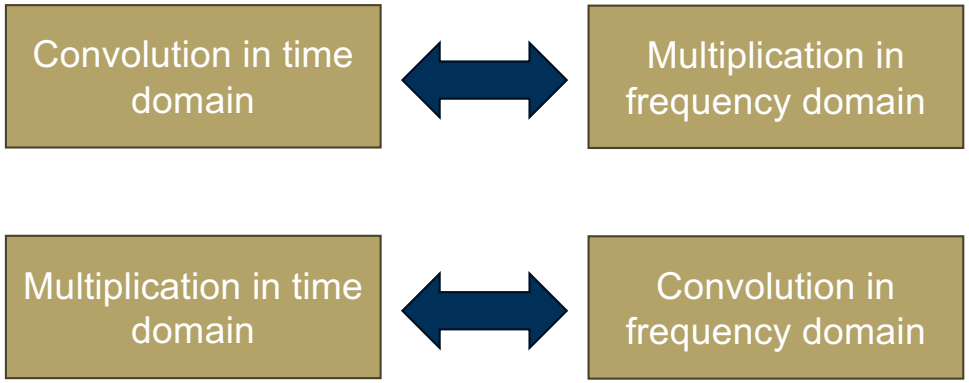
$$(f * g)(t) \stackrel{\text{def}}{=} \int_{-\infty}^{\infty} f(\tau)\, g(t-\tau)\, d\tau$$





“Fast” Convolution

- We can take the Fourier Transform of a signal and apply it then multiply in the frequency domain to compute the convolution
- Convolution in the time domain is the same as multiplication in the frequency domain
- Convolution in the frequency domain is the same as multiplication in in the time domain



Applications of the Fourier Transform

- Perceptual Encoding --> MP3s
- Pitch/Chord Detection
- Noise Cancellation
- Better/Informed Filtering

Non-Audio

- Image compression --> JPEGs
- Heat Flow – Partial Differential Equations
- Earthquakes
- Bridges